

VisionEntropy

Software Architecture
and
Research Roadmap

Version 0.1

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Part I

Motivation and Scientific Vision

Chapter 1

Introduction

1.1 Motivation

Entropy has been one of the fundamental concepts in information theory for more than seventy years. Since Shannon's pioneering work, entropy has become a standard measure of information, uncertainty, complexity, and randomness.

During the last decades entropy has found numerous applications in image processing, including

- image thresholding,
- image segmentation,
- texture analysis,
- image registration,
- feature selection,
- image compression,
- medical imaging,
- remote sensing,
- object recognition.

Most existing software libraries, however, implement only a very limited subset of entropy measures.

Typically they provide

- Shannon entropy,
- histogram entropy,
- local entropy,
- mutual information,
- cross entropy.

Modern computer vision has evolved considerably beyond these classical methods. Today images are represented by

- superpixels,
- graphs,
- convolutional feature maps,
- transformer tokens,
- latent embeddings,
- foundation-model representations.

Similarly, uncertainty is no longer described solely by probability. Current research also employs

- fuzzy sets,
- rough sets,
- fuzzy-rough systems,
- evidence theory,
- Bayesian uncertainty,
- deep learned uncertainty.

Consequently, a modern entropy software framework should support all these representations within a unified architecture.

1.2 Vision

The long-term objective of the VisionEntropy project is to become a comprehensive open-source research platform for entropy-based image analysis.

Rather than implementing isolated entropy measures, the project aims to provide a coherent ecosystem supporting classical image processing, graph analysis, uncertainty modelling, and modern deep learning.

The project is designed around five major principles.

1. Comprehensive implementation of entropy measures.
2. Unified software architecture.
3. Modern deep learning integration.
4. Reproducible scientific experiments.
5. Extensibility for future research.

1.3 Scientific Goals

The project has several scientific objectives.

Goal 1

Create the most comprehensive open-source implementation of entropy methods for image processing.

This includes

- Shannon entropy,
- Rényi entropy,
- Tsallis entropy,
- Approximate entropy,
- Sample entropy,
- Permutation entropy,
- Graph entropy,
- Structural entropy,
- Spectral entropy,
- Fuzzy entropy,
- Rough entropy,
- Fuzzy-rough entropy,
- Feature entropy,
- Latent entropy,
- Predictive entropy.

Goal 2

Provide implementations for both classical image processing and modern deep learning.

The software should work equally well with

- grayscale images,
- RGB images,
- multispectral images,
- hyperspectral images,
- convolutional neural networks,
- Vision Transformers,
- foundation models.

Goal 3

Provide reusable implementations suitable for scientific research.

Every algorithm should include

- mathematical definition,
- Python implementation,
- documentation,
- visualization,
- unit tests,
- benchmark examples.

Goal 4

Support research into entirely new entropy formulations.

Examples include

- Deep Rough Entropy,
- Graph Rough Entropy,
- Fuzzy-Rough Feature Entropy,
- Multi-scale Entropy Learning,
- Entropy-guided Segmentation,
- Entropy-guided Representation Learning.

1.4 Research Philosophy

The software is intended to bridge several traditionally separate research communities.

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box/.style= draw, rounded corners, minimum width=4cm, minimum height=9mm,
           fill=blue!6 ,
           arrow/.style= -Latex, thick
]
[box] (a) Information Theory;
[box,right=2cm of a] (b) Computer Vision;
[box,below=1.3cm of a] (c) Rough Sets;
[box,right=2cm of c] (d) Deep Learning;
[arrow] (a)–(b);
[arrow] (a)–(c);
[arrow] (b)–(d);
[arrow] (c)–(d);

```

Instead of viewing these fields independently, VisionEntropy aims to combine their methods into one consistent computational framework.

1.5 Target Users

The software targets several communities.

- Researchers in computer vision.
- Machine learning researchers.
- Image processing engineers.
- Medical imaging researchers.
- Remote sensing researchers.
- PhD students.
- Graduate students.
- Open-source developers.

1.6 Long-Term Objectives

The project is envisioned as a long-term research platform.

Future versions should include

1. Complete entropy library.
2. Large benchmark collection.
3. Deep-learning toolkit.
4. Graph analysis framework.
5. Differentiable entropy losses.
6. Foundation-model support.
7. Interactive visualization tools.
8. Educational notebooks.
9. Research reproducibility framework.
10. Community contributions.

1.7 Roadmap Structure

The remainder of this roadmap is divided into several major parts.

1. Scientific Background
2. Software Architecture
3. Entropy Algorithms
4. Datasets
5. Deep Learning

6. Experimental Framework
7. Development Roadmap
8. Future Research

Chapter 2

Scientific Background

The VisionEntropy project is based on three fundamental observations.

2.1 Observation 1

Entropy has evolved from pixel statistics toward semantic image representations.

2.2 Observation 2

Uncertainty models have evolved from probability toward fuzzy, rough and learned uncertainty.

2.3 Observation 3

Modern deep learning enables entropy to become an optimization objective, rather than merely a descriptive statistic.

These observations motivate the unified software architecture developed in subsequent chapters.